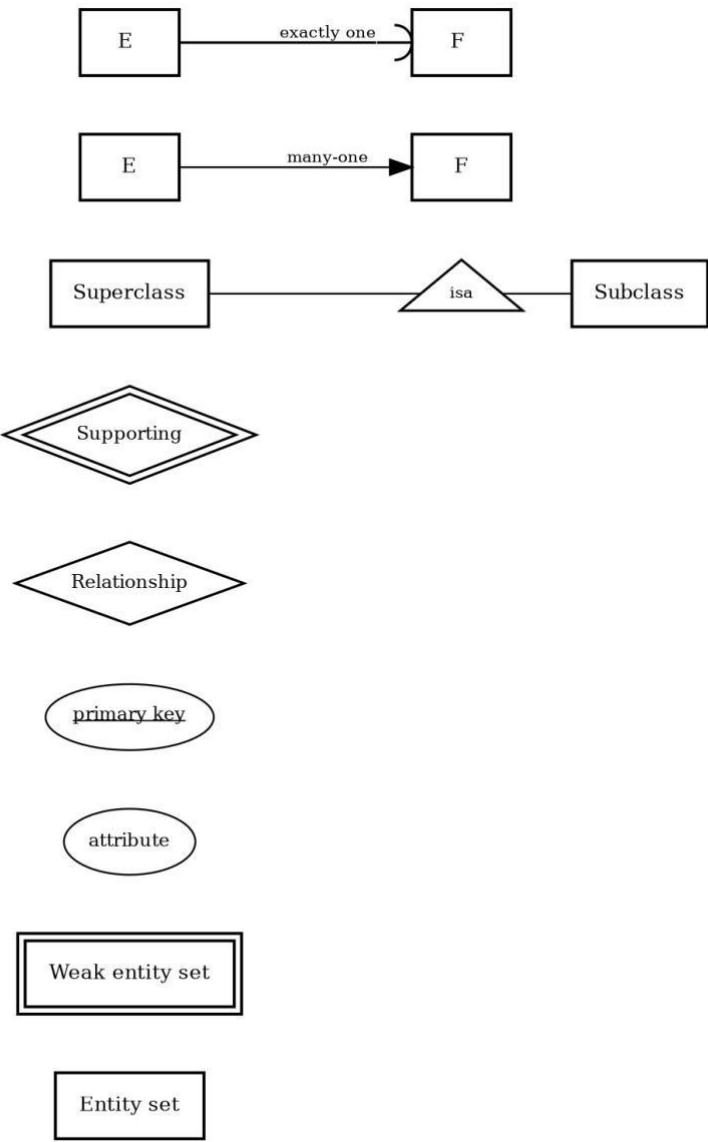
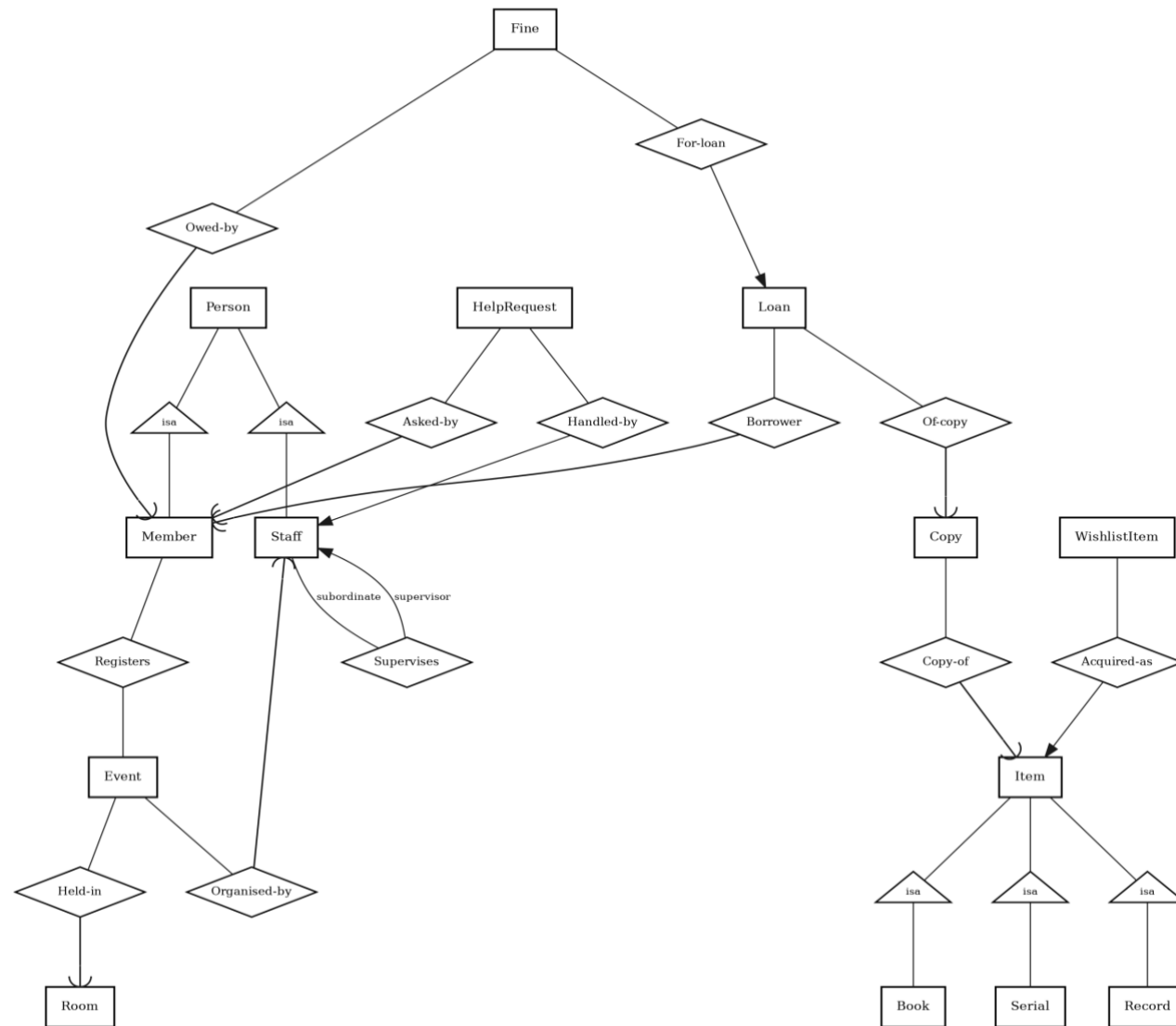


E/R notation used



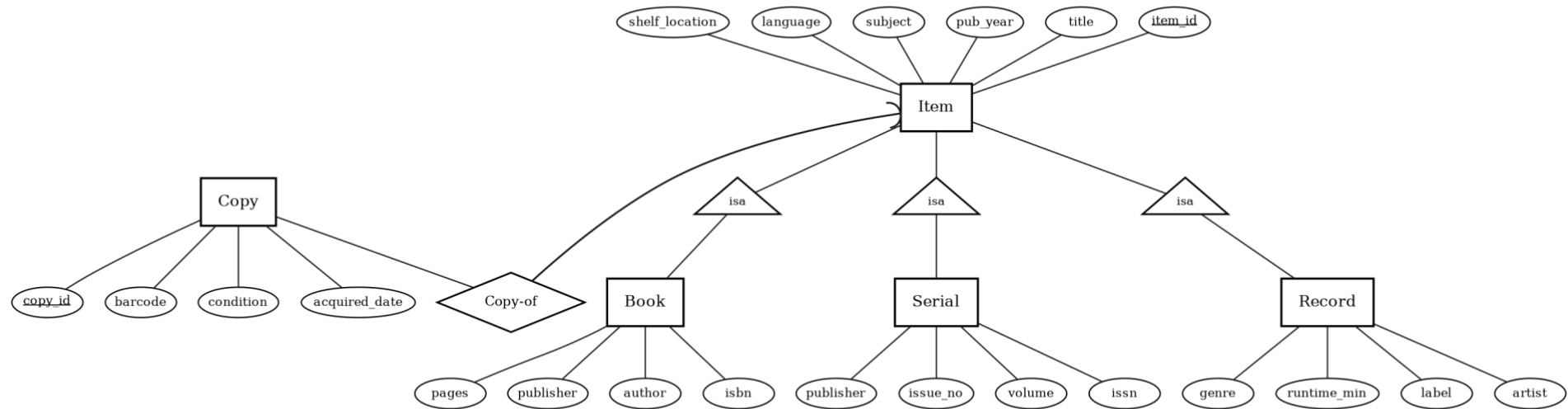
Overview - all entity sets and relationships



15 relations. Attributes omitted for legibility; each cluster is expanded on the following pages.

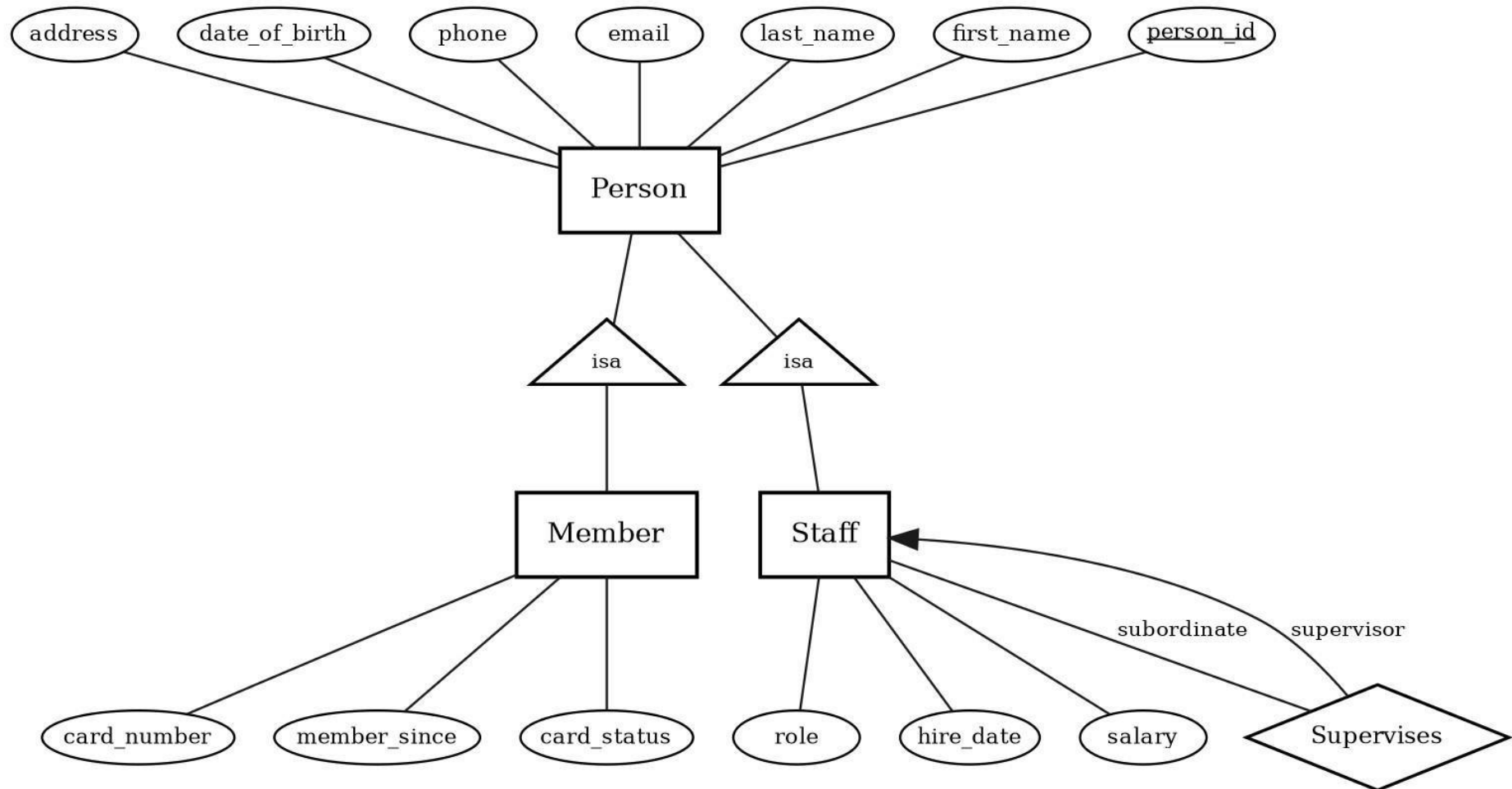
1.

Library items and copies



Item is specialised by content type. *Copy* is a strong entity set: one row per physical object, so availability is derived from *Loan* rather than stored. *shelf_location* sits on *Item* because all copies of a title are shelved together.

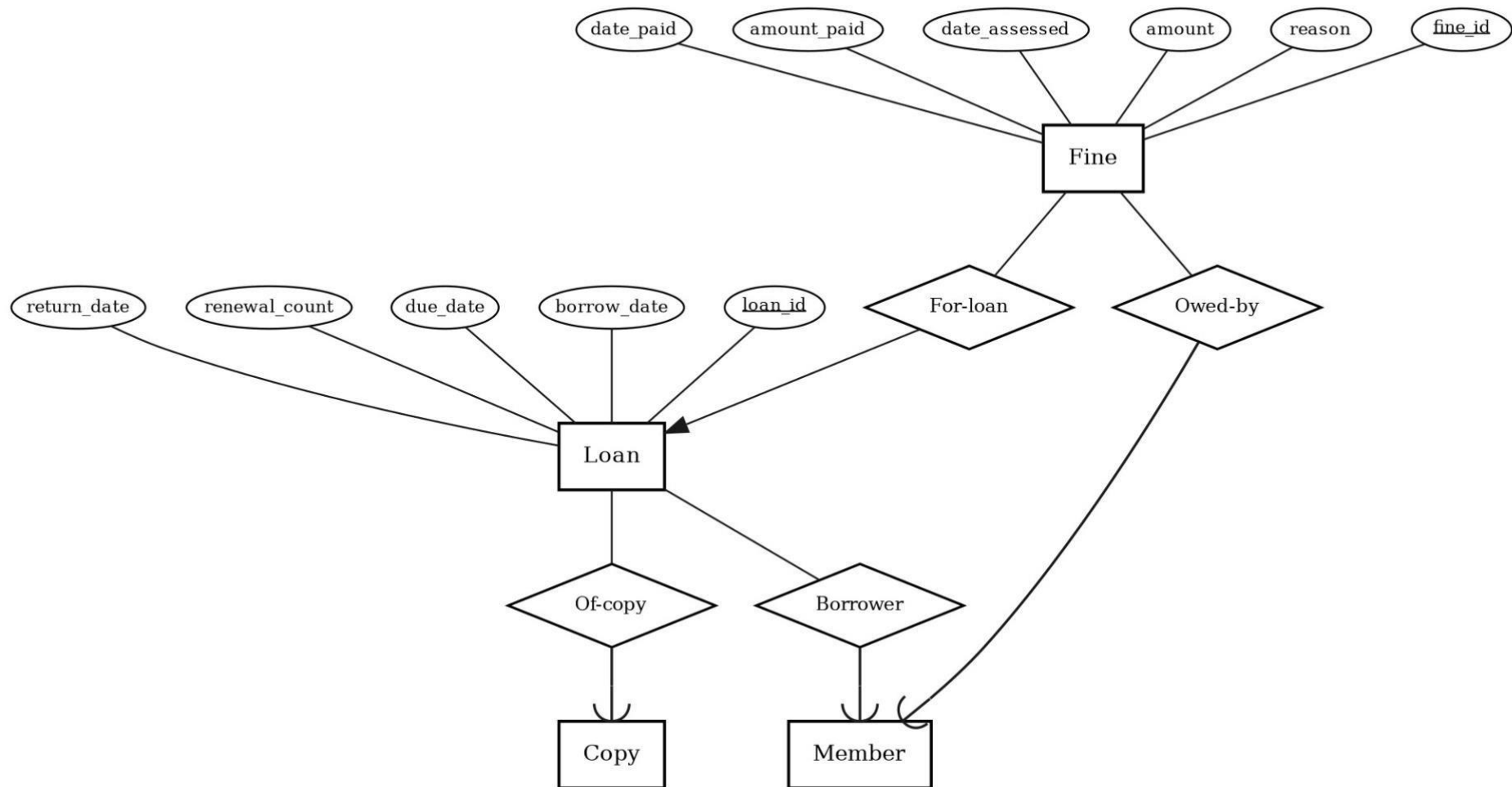
2.

People and personnel

3.

Overlapping specialisation: a staff member may also hold a library card. Supervises is a recursive many-one relationship with a role label on each edge.

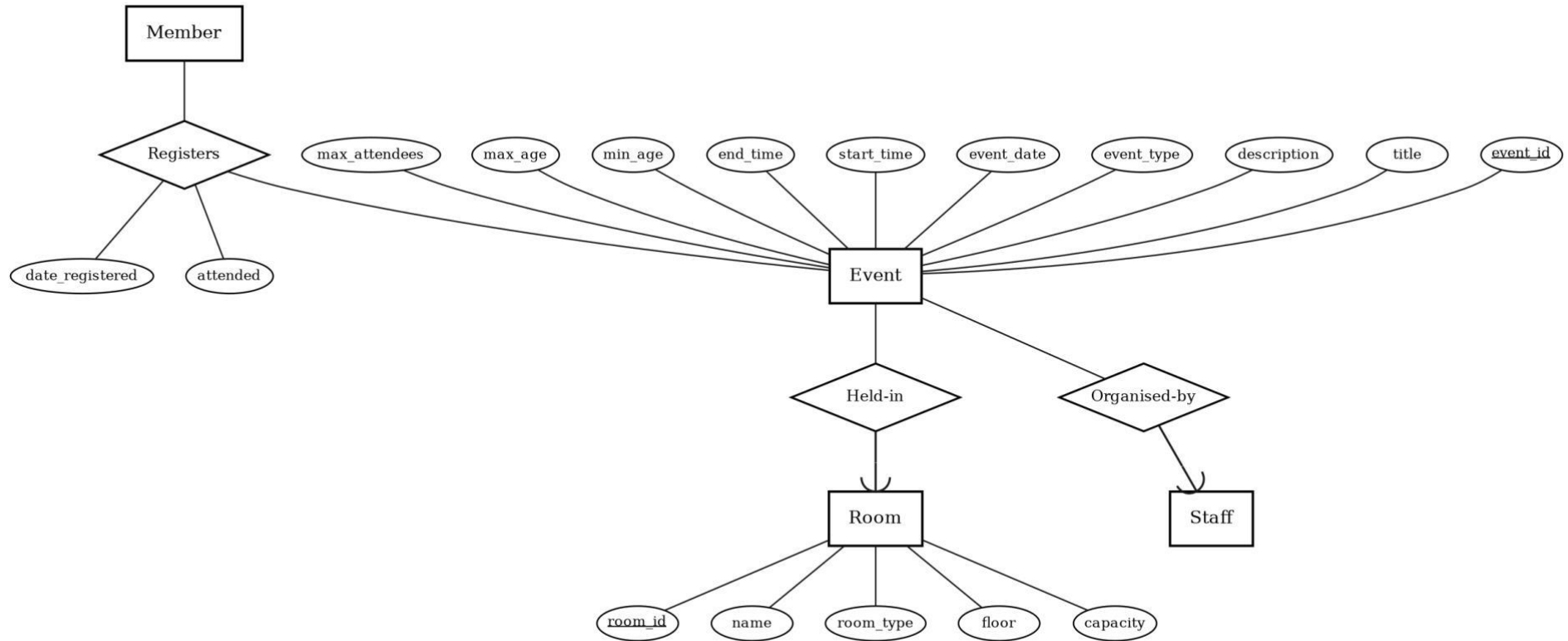
Circulation: loans and fines



4.

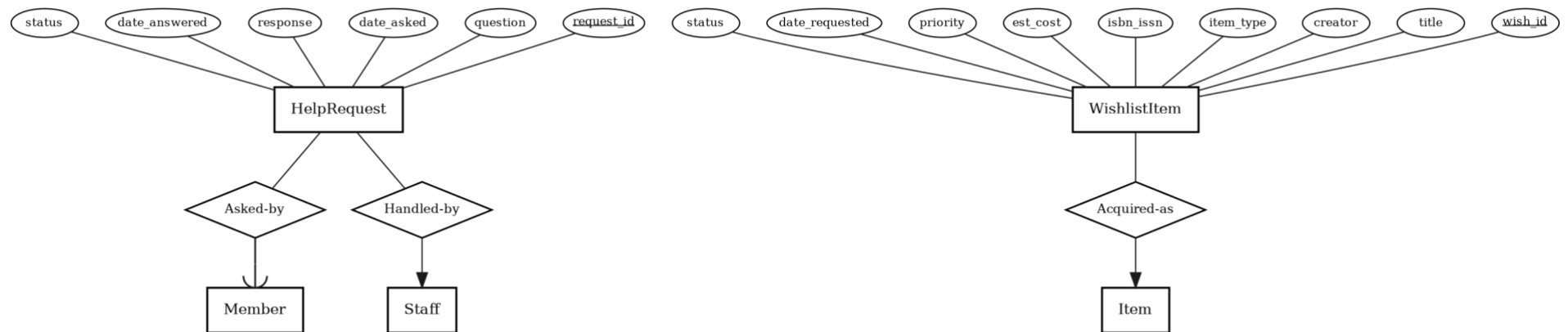
Loan is an entity set, not a relationship, because the same member may borrow the same copy on separate occasions. renewal_count makes due_date independent of borrow_date. For-loan is partial: a fine need not arise from a loan.

Events and rooms



Registers is a many-to-many relationship carrying its own attributes. min_age/max_age are set per event, not per event_type. Room.capacity is the physical limit; Event.max_attendees is the registration cap.

5. Reference services and acquisitions



Handled-by is partial: a request is queued before a librarian picks it up. WishlistItem records intended acquisitions and holds no copies; Acquired-as links it to the Item eventually catalogued.

6. Final Schema derived from the E/R diagrams (15 relations)

Item(item_id, title, pub_year, subject, language, shelf_location)

Book(item_id, isbn, author, publisher, pages)

Serial(item_id, issn, volume, issue_no, publisher)

Record(item_id, artist, label, runtime_min, genre)

Copy(copy_id, item_id, barcode, condition, acquired_date)

Person(person_id, first_name, last_name, email, phone, date_of_birth, address)

Member(person_id, card_number, member_since, card_status)

Staff(person_id, role, hire_date, salary, supervisor_id)

Loan(loan_id, copy_id, member_id, borrow_date, due_date, renewal_count, return_date)

Fine(fine_id, member_id, loan_id, reason, amount, date_assessed, amount_paid, date_paid)

Room(room_id, name, room_type, floor, capacity)

Event(event_id, title, description, event_type, event_date, start_time, end_time, min_age, max_age, max_attendees, room_id, organizer_id)

EventRegistration(event_id, member_id, date_registered, attended)

HelpRequest(request_id, member_id, staff_id, question, date_asked, response, date_answered, status)

WishlistItem(wish_id, title, creator, item_type, isbn_issn, est_cost, priority, date_requested, status, acquired_item)